

Risk Analytics and Claims Intelligence at AegisLife Insurance Pvt. Ltd.

Business Context

AegisLife Insurance Pvt. Ltd., headquartered in Mumbai, is a mid-sized Indian life and general insurance company that has recorded a consistent 35% year-on-year growth in policy issuance over the last three years. This rapid expansion has brought new operational complexities, especially in how customer, policy, and claims data are monitored and interpreted.

Despite having over **2 million customer records** and a workforce of **1500+ agents** across **six regions**, the company's management is struggling with:

1. **High claim ratios** that threaten profitability across certain regions and product lines
2. **Potential anomalies in claims data**, possibly indicating inefficient or fraudulent processing
3. **Lack of unified, real-time business dashboards** for decision-makers to track performance metrics

The executive team believes that a **comprehensive data analytics initiative** can help them uncover actionable insights. By leveraging tools like **SQL, Python, Excel, Statistics, and Power BI**, AegisLife aims to build a robust **analytics foundation** to improve operational oversight, ensure compliance, and enhance financial reporting.

Project Objective

You have been onboarded as a **Data Analyst Consultant** for the Analytics Center of Excellence (ACoE) at *AegisLife Insurance Pvt. Ltd.* Your responsibility is to analyze large volumes of operational and claims-related data to support data-driven decision making across key insurance processes.

Your work will focus on:

- **Building a comprehensive data analytics workflow** using SQL, Python, and Excel
- **Conducting statistical analysis** to validate business hypotheses related to claims, customer risk, and agent performance
- **Developing interactive dashboards in Power BI** to support real-time monitoring and executive reporting
- **Delivering actionable business insights** with quantifiable financial impact and strategic recommendations

Suggested Project Scope & Tools

Tool	Applications
Excel	Exploratory summaries, scenario modeling, hypothesis tracking
SQL	Data extraction, joins, aggregations from relational databases
Python	Data cleaning, transformation, EDA, Data wrangling
Statistics	Inferential testing, distributions, probability modeling
Power BI / Tableau	Executive dashboard with KPIs, visual analytics, and filters

Project Deliverables

1. Business Understanding & KPI Framework
 - Translate problem into analytical objectives
 - Define and document key insurance KPIs, for example:
Loss ratio, Claim frequency, Agent productivity, Policy lapse rate
 - Segment business challenges across region, product type, and agent tier
2. Data Acquisition & Cleaning
 - Extract and merge data using SQL queries
 - Perform data cleaning and wrangling in Python and Excel
(e.g., missing values, inconsistent formats, outliers)
 - Validate data integrity across linked tables (customer-policy-claim-agent)
3. Exploratory Analysis & Descriptive Statistics
 - Generate frequency distributions, cross-tabs, and pivots (in Excel & Python)
 - Analyze trends: claim volume over time, high-claim product lines, agent-wise performance
 - Compute descriptive statistics (mean, median, std. dev) by key dimensions
4. Statistical Testing & Business Hypothesis Validation
 - Perform t-tests / ANOVA to compare claim behavior across age groups, regions, policy types
 - Use probability modeling to estimate risks (e.g., smoker vs non-smoker claim rates)
 - Test business assumptions with significance tests and confidence intervals
5. Data Storytelling & Dashboarding with Power BI or Tableau
 - Design and publish an interactive executive dashboard:
 - KPIs: Total claims, avg. claim amount, claim approval rate, loss ratio
 - Filters: Region, product type, agent, time period
 - Charts: Trend lines, heatmaps, bar charts, donut charts, tables
 - Create role-specific views: Executive vs Regional Manager

6. Strategic Insights & Recommendations

- Compile a concise executive summary report (2 pages):
- Key insights from data exploration and hypothesis tests
- Region-wise or product-wise performance flags
- Data-driven recommendations for operations, monitoring, and compliance

Data Tables Provided

- Customer_Master

Column Name	Data Type	Description
Customer_ID	String	Unique identifier (CUSTxxxxx)
Full_Name	String	Customer name
Age	Integer	Age in years
Gender	Categorical	Male / Female / Other
Marital_Status	Categorical	Single / Married / Divorced
Occupation	Categorical	Salaried / Self-employed / Retired / Student
Region	Categorical	North / South / East / West / Central / North-East
Smoking_Status	Boolean	Yes / No
Pre_Existing_Illness	Boolean	Yes / No
Risk_Score	Float	Calculated score from 0–1
Date_Joined	Date	Date of first policy

- Policy_Details

Column Name	Data Type	Description
Policy_ID	String	Unique policy number (e.g., POL500001)
Customer_ID	String	FK → Customer_Master.Customer_ID
Product_Type	Categorical	Term / Whole / Health / Vehicle / Property
Coverage_Amount	Float	Sum insured (INR)
Annual_Premium	Float	Premium paid annually (INR)
Policy_Start_Date	Date	Start date of coverage
Policy_End_Date	Date	End date of coverage
Agent_ID	String	FK → Agent_Info.Agent_ID
Status	Categorical	Active / Lapsed / Cancelled

- Claim_History

Column Name	Data Type	Description
Claim_ID	String	Unique claim reference number
Policy_ID	String	FK → Policy_Details.Policy_ID
Claim_Date	Date	Date claim was filed
Claim_Amount	Float	Amount claimed (INR)
Claim_Status	Categorical	Approved / Rejected / Pending
Claim_Type	Categorical	Hospital / Accident / Death / Theft / Other
Fraud_Flag	Boolean	Yes / No
Days_To_Process	Integer	Processing time in days

- Agent_Info

Column Name	Data Type	Description
Agent_ID	String	Unique agent identifier
Region	Categorical	Region of operation
Join_Date	Date	Date of onboarding

Column Name	Data Type	Description
Total_Policies_Sold	Integer	Number of policies sold
Lapsed_Policies	Integer	Count of policies that lapsed
Avg_Premium_Sold	Float	Average premium value per policy
Fraud_Association	Integer	Count of fraud-flagged claims sold under this agent

- Customer_Feedback_Surveys

Column Name	Data Type	Description
Feedback_ID	String	Unique feedback record
Customer_ID	String	FK → Customer_Master.Customer_ID
Date_Submitted	Date	Date of feedback
Feedback_Text	String	Textual feedback (open-ended)
Satisfaction_Score	Integer	Rating from 1 (low) to 5 (high)
Contacted_Agent	Boolean	Yes / No
Referred_Claim	Boolean	Whether feedback relates to a claim

Primary Key Foreign Key Relationships

Table	Foreign Key	References
policy_details	customer_id	customer_master.customer_id
policy_details	agent_id	agent_info.agent_id
claim_history	policy_id	policy_details.policy_id
customer_feedback_surveys	customer_id	customer_master.customer_id

SUGGESTED INSTRUCTIONS

Learners are encouraged to apply their own ideas while executing the project. The following is intended solely as a guideline.

Note: Learners are supposed to check data integrity and handle any inconsistencies accordingly.

Hint: Customer feedback may be given by customers without disclosing identity.

Step 1: Business Understanding & Analytical Planning

Objective: Develop a clear understanding of business context and define analytical goals

- Start with a **written note (preferably PowerPoint/Google Slide)** outlining:
 - The core business challenges of AegisLife Insurance.
 - Key analytical goals from a business intelligence perspective.
 - Identify the **core KPIs** like Claim Frequency, Loss Ratio, Agent Productivity, and Policy Lapse Rate. You can add more as per your analysis.
 - Formulate hypotheses related to these KPIs (e.g., "Agents in the West region have higher fraud association than the national average").
 - Present this summary as a brief write-up in PowerPoint/Google Slide.

Step 2: Data Validation Using Excel

Objective: Validate data integrity and address basic inconsistencies before ingestion

- Use techniques to verify foreign key relationships for example:
 - Ensure all Customer_IDs in Policy_Details exist in Customer_Master
- Apply **conditional formatting** to highlight:
 - Missing or blank fields in critical columns such as Age, Risk_Score, Claim_Amount, and Claim_Status
 - Unexpected or inconsistent categorical entries (e.g., typos in Claim_Status)
- Perform **basic data cleaning** in Excel, if required:
 - Standardize categorical values (e.g., correcting "mal" to "Male")
 - Format date columns consistently across all tables
 - Flag missing values that require imputation later (avoid deleting at this stage)
 - Create a flag column like Validation_Status to tag rows needing review
- Create a **"Data Quality Log" worksheet** to summarize:
 - Column names with detected issues
 - Type of inconsistency (e.g., missing, invalid, outlier)
 - Suggested cleaning action
 - Count or percentage of affected rows
- Summarize your analysis:

- Use PowerPoint/Google Slide

Step 3: Database Design and Table Creation (SQL DDL)

Objective: Design and structure the relational database schema

- Write SQL Data Definition Language (DDL) scripts to create tables for all datasets:
 - Define correct data types, primary keys, and foreign key constraints
 - Include constraints wherever applicable
- Save all DDL scripts in a text file and validate execution in your SQL environment.

Step 4: Data Ingestion into SQL

Objective: Import cleaned data into a structured relational database

- Load the Excel datasets into the respective SQL tables:
 - Use native import features or SQL queries like LOAD DATA INFILE, or tools like MySQL Workbench
 - Optionally, use Python to automate this process
- Post-ingestion, validate:
 - Row counts across tables
 - Random spot checks for correct field mapping
 - Consistency of foreign key relationships after loading

Step 5: Connecting Python to SQL

Objective: Establish a dynamic data pipeline from SQL to Python for analysis

- In a Jupyter Notebook, demonstrate how to:
 - Install and import required libraries (pymysql, sqlalchemy, pandas)
 - Construct a secure connection string to your SQL database
 - Execute SQL queries from Python using pandas.read_sql_query
 - Load data from all six tables into Pandas dataframes for further use
- Validate schema and record counts in Python to ensure successful extraction.

Step 6: Data Cleaning, Transformations & Enhanced Exploratory Data Analysis (EDA) in Python

Objective: Prepare data for analysis and uncover initial trends

- Clean the extracted data:
 - Treat missing values through imputation or removal
 - Normalize and standardize numerical fields like Risk_Score, Claim_Amount. Interpret.
 - Create derived columns. Below are examples only:
 - Policy Duration = Policy_End_Date - Policy_Start_Date
 - Claims per Customer

- $\text{Agent Fraud Ratio} = \text{Fraud_Association} / \text{Total_Policies_Sold}$
- Perform EDA using pandas, matplotlib, and seaborn:
 - Visualize trends over time (e.g., monthly claim volumes)
 - Use heatmaps and boxplots to detect regional and product-level patterns
 - Summarize metrics using grouped statistics (mean, median, standard deviation)

Step 7: Statistical Analysis in Python

Objective: Validate business hypotheses using statistical tests

- Perform statistical tests using scipy.stats and statsmodels. The following are examples only
 - T-tests to compare average claim amounts across customer types (e.g., smoker vs non-smoker)
 - ANOVA to evaluate claim variability across regions
 - Chi-square test for categorical variable associations (e.g., Claim_Status vs Claim_Type)
 - Correlation analysis between Risk_Score and Claim_Amount
- Interpret test outputs. The following are examples only.
 - Include p-values, confidence intervals, and visualizations (e.g., boxplots, histograms)
 - Document test assumptions and limitations

Step 8: Power BI / Tableau Dashboard Development

Objective: Build executive dashboards for performance monitoring and decision support

- Connect Power BI Desktop or Tableau to the MySQL database
- Build an integrated data model using all relevant tables
- Create the following dashboards. The following are examples only.
 - **Executive Dashboard:**
 - KPIs: Total Claims, Avg Claim Amount, Approval Rate, Loss Ratio, Fraud Ratios
 - Visuals: Line charts for trends, donut charts for product types, bar graphs for agent comparison
 - Filters: Region, Product Type, Claim Status, Time Period
 - **Regional Manager Dashboard:**
 - Drill-down views by region
 - Specific agent and customer performance metrics
- Ensure dashboards are clean, interactive, and easy to interpret for non-technical audiences

Step 9: Strategic Insights & Recommendations

Objective: Derive actionable business insights and communicate them effectively. The following are examples only.

- Draft a concise 2-page executive report covering.
 - Top 5 data-driven insights (e.g., agents with unusually high claim rejection, underperforming product categories)
 - Supporting visualizations from Power BI/Tableau or Python
 - At least 3 strategic recommendations (e.g., targeted agent training, policy redesign, fraud investigation focus)
- Ensure recommendations are backed by metrics and include potential business impact estimates